



National  
Qualifications  
2019

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# **2019 Mathematics**

## **Higher Paper 2**

### **Finalised Marking Instructions**

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## General marking principles for Higher Mathematics

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

*For each question, the marking instructions are generally in two sections:*

- *generic scheme – this indicates why each mark is awarded*
- *illustrative scheme – this covers methods which are commonly seen throughout the marking*

*In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.*

- Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.
- One mark is available for each •. There are no half marks.
- If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.
- Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.
- Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.
- If an error is trivial, casual or insignificant, for example  $6 \times 6 = 12$ , candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.
- If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 1 \text{ or } 3$$

(i) **Horizontal/vertical marking**

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

$$\begin{array}{cc} \bullet^5 & \bullet^6 \\ \bullet^5 & x = 2 \quad x = -4 \\ \bullet^6 & y = 5 \quad y = -7 \end{array}$$

Horizontal:  $\bullet^5 x = 2$  and  $x = -4$       Vertical:  $\bullet^5 x = 2$  and  $y = 5$   
 $\bullet^6 y = 5$  and  $y = -7$        $\bullet^6 x = -4$  and  $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$$\begin{array}{ll} \frac{15}{12} \text{ must be simplified to } \frac{5}{4} \text{ or } 1\frac{1}{4} & \frac{43}{1} \text{ must be simplified to } 43 \\ \frac{15}{0.3} \text{ must be simplified to } 50 & \frac{4\cancel{5}}{3} \text{ must be simplified to } \frac{4}{15} \\ \sqrt{64} \text{ must be simplified to } 8^* & \end{array}$$

\*The square root of perfect squares up to and including 100 must be known.

(k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

(l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example

$$\begin{aligned} & (x^3 + 2x^2 + 3x + 2)(2x + 1) \text{ written as} \\ & (x^3 + 2x^2 + 3x + 2) \times 2x + 1 \\ & = 2x^4 + 5x^3 + 8x^2 + 7x + 2 \\ & \text{gains full credit} \end{aligned}$$

- repeated error within a question, but not between questions or papers

(m) In any 'Show that...' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

- (n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.
- (o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.
- (p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks.                             | Strategy 2 attempt 1 is worth 1 mark.                              |
| Strategy 1 attempt 2 is worth 4 marks.                             | Strategy 2 attempt 2 is worth 5 marks.                             |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

| Question                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | Generic scheme                                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                     | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.                                                                                                                                                                                                                                                                                                                                                                                                          | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> calculate the midpoint of AC</li> <li>•<sup>2</sup> calculate the gradient of BD</li> <li>•<sup>3</sup> determine equation of BD</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(-4, -3)</math></li> <li>•<sup>2</sup> <math>-\frac{1}{3}</math></li> <li>•<sup>3</sup> <math>3y = -x - 13</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                               |     |  |                                                                                                                                                                                                  |                                                                                                                                                                                         |          |
| 1. • <sup>2</sup> is only available to candidates who use a midpoint to find a gradient.<br>2. • <sup>3</sup> is only available as a consequence of using the midpoint of AC and the point B.<br>3. At • <sup>3</sup> accept any arrangement of a candidate's equation where constant terms have been simplified.<br>4. • <sup>3</sup> is not available as a consequence of using a perpendicular gradient. |     |  |                                                                                                                                                                                                  |                                                                                                                                                                                         |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                         |     |  |                                                                                                                                                                                                  |                                                                                                                                                                                         |          |
| <b>Candidate A</b> - Perpendicular Bisector of AC                                                                                                                                                                                                                                                                                                                                                           |     |  | <b>Candidate B</b> - Altitude through B                                                                                                                                                          |                                                                                                                                                                                         |          |
| Midpoint <sub>AC</sub> $(-4, -3)$                                                                                                                                                                                                                                                                                                                                                                           |     |  | $m_{AC} = 9$                                                                                                                                                                                     |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | • <sup>1</sup> ✓                                                                                                                                                                                 |                                                                                                                                                                                         |          |
| $m_{AC} = 9 \Rightarrow m_{\perp} = -\frac{1}{9}$                                                                                                                                                                                                                                                                                                                                                           |     |  | • <sup>2</sup> ✗                                                                                                                                                                                 |                                                                                                                                                                                         |          |
| $9y + x + 31 = 0$                                                                                                                                                                                                                                                                                                                                                                                           |     |  | • <sup>3</sup> ✓ 2                                                                                                                                                                               |                                                                                                                                                                                         |          |
| For other perpendicular bisectors award 0/3                                                                                                                                                                                                                                                                                                                                                                 |     |  | $9y + x = -61$                                                                                                                                                                                   |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | • <sup>3</sup> ✓ 2                                                                                                                                                                               |                                                                                                                                                                                         |          |
| <b>Candidate C</b> - Median through A                                                                                                                                                                                                                                                                                                                                                                       |     |  | <b>Candidate D</b> - Median through C                                                                                                                                                            |                                                                                                                                                                                         |          |
| Midpoint <sub>BC</sub> $(4, -1)$                                                                                                                                                                                                                                                                                                                                                                            |     |  | Midpoint <sub>AB</sub> $(3, -10)$                                                                                                                                                                |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | • <sup>1</sup> ✗                                                                                                                                                                                 |                                                                                                                                                                                         |          |
| $m_{AM} = \frac{11}{9}$                                                                                                                                                                                                                                                                                                                                                                                     |     |  | • <sup>2</sup> ✓ 1                                                                                                                                                                               |                                                                                                                                                                                         |          |
| $9y - 11x + 53 = 0$                                                                                                                                                                                                                                                                                                                                                                                         |     |  | $m_{CM} = -\frac{8}{3}$                                                                                                                                                                          |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | • <sup>3</sup> ✓ 2                                                                                                                                                                               |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | $3y + 8x + 6 = 0$                                                                                                                                                                                |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | • <sup>3</sup> ✓ 2                                                                                                                                                                               |                                                                                                                                                                                         |          |

| Question                                                                                                                                                                                                         |     |  | Generic scheme                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                        | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                                                                                                                                                                                                  | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> calculate gradient of BC</li> <li>•<sup>5</sup> use property of perpendicular lines</li> <li>•<sup>6</sup> determine equation of AE</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>-1</math></li> <li>•<sup>5</sup> <math>1</math></li> <li>•<sup>6</sup> <math>y = x - 7</math></li> </ul>        | 3        |
| <b>Notes:</b>                                                                                                                                                                                                    |     |  |                                                                                                                                                                                                     |                                                                                                                                                                            |          |
| 5. • <sup>6</sup> is only available to candidates who find and use a perpendicular gradient.<br>6. At • <sup>6</sup> accept any arrangement of a candidate's equation where constant terms have been simplified. |     |  |                                                                                                                                                                                                     |                                                                                                                                                                            |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                              |     |  |                                                                                                                                                                                                     |                                                                                                                                                                            |          |
| <b>Candidate E</b><br>Correct gradient from incorrect substitution<br>$m_{BC} = \frac{-3 - 11}{6 + 8} = -1$<br>$m_{AE} = 1$<br>$y = x - 7$                                                                       |     |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> ✗</li> <li>•<sup>5</sup> <input checked="" type="checkbox"/> 1</li> <li>•<sup>6</sup> <input checked="" type="checkbox"/> 1</li> </ul>         |                                                                                                                                                                            |          |
|                                                                                                                                                                                                                  | (c) |  | <ul style="list-style-type: none"> <li>•<sup>7</sup> find <math>x</math> or <math>y</math> coordinate</li> <li>•<sup>8</sup> find remaining coordinate of the point of intersection</li> </ul>      | <ul style="list-style-type: none"> <li>•<sup>7</sup> <math>x = 2</math> or <math>y = -5</math></li> <li>•<sup>8</sup> <math>y = -5</math> or <math>x = 2</math></li> </ul> | 2        |
| <b>Notes:</b>                                                                                                                                                                                                    |     |  |                                                                                                                                                                                                     |                                                                                                                                                                            |          |
| 7. For $(2, -5)$ with no working, award 0/2.                                                                                                                                                                     |     |  |                                                                                                                                                                                                     |                                                                                                                                                                            |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                              |     |  |                                                                                                                                                                                                     |                                                                                                                                                                            |          |
|                                                                                                                                                                                                                  |     |  |                                                                                                                                                                                                     |                                                                                                                                                                            |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                                                                                                                                                       | Max mark |
|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> express <math>6\sqrt{x}</math> in integrable form</li> <li>•<sup>2</sup> integrate first term</li> <li>•<sup>3</sup> integrate second term</li> <li>•<sup>4</sup> complete integration</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>6x^{\frac{1}{2}}</math></li> <li>•<sup>2</sup> <math>\frac{6x^{\frac{3}{2}}}{\frac{3}{2}} \dots</math></li> <li>•<sup>3</sup> <math>\dots - \frac{4x^{-2}}{-2} \dots</math></li> <li>•<sup>4</sup> <math>4x^{\frac{3}{2}} + 2x^{-2} + 5x + c</math></li> </ul> | 4        |

#### Notes:

1. •<sup>2</sup> is only available for integrating a term with a fractional index.
2. All coefficients must be simplified at •<sup>4</sup> stage for •<sup>4</sup> to be awarded.
3. Do not penalise the appearance of an integral sign throughout.
4. Do not penalise the omission of '+c' at •<sup>2</sup> and •<sup>3</sup>.

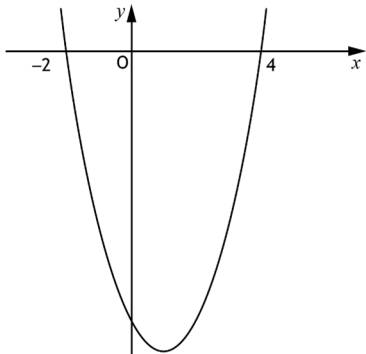
#### Commonly Observed Responses:

##### Candidate A

$$\begin{aligned}
 & \int \left( 6x^{\frac{1}{2}} - 4x^{-3} + 5 \right) dx && \bullet^1 \checkmark \\
 & = \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{4x^{-2}}{-2} + 5x + c && \bullet^2 \checkmark \quad \bullet^3 \checkmark \\
 & = \frac{12}{3} x^{\frac{3}{2}} + 2x^{-2} + 5x + c && \\
 & = 4\sqrt{x^3} + \frac{2}{\sqrt{x}} + 5x + c && \bullet^4 \times \\
 & \bullet^4 \text{ cannot be awarded over two lines of working}
 \end{aligned}$$

| Question                                                                                                                                    |     |  | Generic scheme                                                                                                                                                                              | Illustrative scheme                                                                                                                                                                                             | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3.                                                                                                                                          | (a) |  | • <sup>1</sup> identify pathway                                                                                                                                                             | • <sup>1</sup> $-\mathbf{p} + \mathbf{r}$                                                                                                                                                                       | 1        |
| Notes:                                                                                                                                      |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                 |          |
| 1. Accept $-\mathbf{P} + \mathbf{R}$ for • <sup>1</sup> .                                                                                   |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                 |          |
| Commonly Observed Responses:                                                                                                                |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                 |          |
|                                                                                                                                             |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                 |          |
|                                                                                                                                             | (b) |  | • <sup>2</sup> state an appropriate pathway<br><br>• <sup>3</sup> express pathway in terms of $\mathbf{p}$ , $\mathbf{q}$ and $\mathbf{r}$                                                  | • <sup>2</sup> eg $\overline{\mathbf{E}\mathbf{B}} + \overline{\mathbf{B}\mathbf{F}}$ stated or implied by • <sup>3</sup><br><br>• <sup>3</sup> $\mathbf{p} - \mathbf{r} + \frac{3}{4}\mathbf{q}$ or equivalent | 2        |
| Notes:                                                                                                                                      |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                 |          |
| 2. • <sup>3</sup> can only be awarded for a vector expressed in terms of all three of $\mathbf{p}$ , $\mathbf{q}$ and $\mathbf{r}$ .        |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                 |          |
| Commonly Observed Responses:                                                                                                                |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                 |          |
| Candidate A - incorrect expression in $\mathbf{p}$ , $\mathbf{q}$ and $\mathbf{r}$ and no pathway stated<br>$\mathbf{p} - \mathbf{r} \dots$ |     |  | Candidate B - incorrect expression in $\mathbf{p}$ , $\mathbf{q}$ and $\mathbf{r}$ and no pathway stated<br>$\dots + \frac{3}{4}\mathbf{q}$ or $\dots + \mathbf{q} - \frac{1}{4}\mathbf{q}$ |                                                                                                                                                                                                                 |          |
| Award 1/2                                                                                                                                   |     |  | Award 1/2                                                                                                                                                                                   |                                                                                                                                                                                                                 |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |      | Generic scheme                                                                                           | Illustrative scheme                                                                      | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------|
| 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (a) |      | • <sup>1</sup> state values of $a$ and $b$                                                               | • <sup>1</sup> $a = 0.973$ , $b = 30$                                                    | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |      |                                                                                                          |                                                                                          |          |
| 1. Accept $u_{n+1} = 0.973u_n + 30$ for • <sup>1</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |      |                                                                                                          |                                                                                          |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      |                                                                                                          |                                                                                          |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (b) | (i)  | • <sup>2</sup> communicate condition for limit to exist                                                  | • <sup>2</sup> a limit exists as the recurrence relation is linear and $-1 < 0.973 < 1$  | 1        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     | (ii) | • <sup>3</sup> know how to find limit<br><br>• <sup>4</sup> process limit and state estimated population | • <sup>3</sup> $L = 0.973L + 30$ or<br>$L = \frac{30}{1 - 0.973}$<br>• <sup>4</sup> 1100 | 2        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |      |                                                                                                          |                                                                                          |          |
| 2. For • <sup>2</sup> accept:<br>$-1 < 0.973 < 1$ or $ 0.973  < 1$ or $0 < 0.973 < 1$ with no further comment;<br>or statements such as “0.973 lies between -1 and 1”;<br>or $-1 < a < 1$ (as $a$ is previously defined).<br>3. • <sup>2</sup> is not available for:<br>$-1 \leq 0.973 \leq 1$ or $0.973 < 1$ ;<br>or statements such as “it is between -1 and 1”<br>4. Do not accept $L = \frac{b}{1-a}$ with no further working for • <sup>3</sup> .<br>5. For $L = 1100$ with no working award • <sup>3</sup> and • <sup>4</sup> . |     |      |                                                                                                          |                                                                                          |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      |                                                                                                          |                                                                                          |          |
| Candidate A - no rounding required                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |      | Candidate B - correct rounding                                                                           |                                                                                          |          |
| $u_{n+1} = 0.97u_n + 30$<br>$\vdots$<br>$L = \frac{30}{1 - 0.97}$<br>$L = 1000$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |      | $u_{n+1} = 0.027u_n + 30$<br>$\vdots$<br>$L = \frac{30}{1 - 0.027}$<br>$L = 0$                           |                                                                                          |          |
| • <sup>1</sup> ✗<br><br>• <sup>3</sup> ✓ 1<br>• <sup>4</sup> ✓ 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |      | • <sup>1</sup> ✗<br><br>• <sup>3</sup> ✓ 1<br>• <sup>4</sup> ✓ 1                                         |                                                                                          |          |
| Candidate C - no valid limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      |                                                                                                          |                                                                                          |          |
| $u_{n+1} = 2.7u_n + 30$<br>A limit does not exist as $2.7 > 1$<br>$L = \frac{30}{1 - 2.7}$<br>$L = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |      |                                                                                                          |                                                                                          |          |
| • <sup>1</sup> ✗<br>• <sup>2</sup> ✗<br>• <sup>3</sup> ✓ 1<br>• <sup>4</sup> ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |      |                                                                                                          |                                                                                          |          |

| Question                                                                                     |  |  | Generic scheme                                                                                                                  | Illustrative scheme                                                                                                                                                                                                                                                                           | Max mark |
|----------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5.                                                                                           |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> identify shape and roots</li> <li>•<sup>2</sup> interpret shape</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> parabola with roots at <math>-2</math> and <math>4</math></li> <li>•<sup>2</sup> parabola with a minimum turning point at <math>x=1</math></li> </ul>  | 2        |
| <b>Notes:</b>                                                                                |  |  |                                                                                                                                 |                                                                                                                                                                                                                                                                                               |          |
| 1. • <sup>1</sup> and • <sup>2</sup> are only available for attempting to draw a 'parabola'. |  |  |                                                                                                                                 |                                                                                                                                                                                                                                                                                               |          |
| <b>Commonly Observed Responses:</b>                                                          |  |  |                                                                                                                                 |                                                                                                                                                                                                                                                                                               |          |
|                                                                                              |  |  |                                                                                                                                 |                                                                                                                                                                                                                                                                                               |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                          | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                        | Max mark |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.       | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use compound angle formula</li> <li>•<sup>2</sup> compare coefficients</li> <li>•<sup>3</sup> process for <math>k</math></li> <li>•<sup>4</sup> process for <math>a</math> and express in required form</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>k \cos x^\circ \cos a^\circ - k \sin x^\circ \sin a^\circ</math><br/>stated explicitly</li> <li>•<sup>2</sup> <math>k \cos a^\circ = 2, k \sin a^\circ = 3</math><br/>stated explicitly</li> <li>•<sup>3</sup> <math>\sqrt{13}</math></li> <li>•<sup>4</sup> <math>\sqrt{13} \cos(x + 56 \cdot 3 \dots)^\circ</math></li> </ul> | 4        |

#### Notes:

- Accept  $k(\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ)$  for •<sup>1</sup>.  
Treat  $k \cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ$  as bad form only if the equations at the •<sup>2</sup> stage both contain  $k$ .
- Do not penalise the omission of degree signs.
- $\sqrt{13} \cos x^\circ \cos a^\circ - \sqrt{13} \sin x^\circ \sin a^\circ$  or  $\sqrt{13}(\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ)$  is acceptable for •<sup>1</sup> and •<sup>3</sup>.
- <sup>2</sup> is not available for  $k \cos x^\circ = 2, k \sin x^\circ = 3$ , however •<sup>4</sup> may still be gained. See Candidate F.
- Accept  $k \cos a^\circ = 2, -k \sin a^\circ = -3$  for •<sup>2</sup>.
- <sup>3</sup> is only available for a single value of  $k, k > 0$ .
- <sup>4</sup> is not available for a value of  $a$  given in radians.
- Accept values of  $a$  which round to 56.
- Candidates may use any form of the wave function for •<sup>1</sup>, •<sup>2</sup> and •<sup>3</sup>.  
However, •<sup>4</sup> is only available if the wave is interpreted in the form  $k \cos(x + a)^\circ$ .
- Evidence for •<sup>4</sup> may not appear until part (b).

#### Commonly Observed Responses:

| Candidate A                                                                                                                                                        | Candidate B                                                                                                                                                                                                   | Candidate C                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\sqrt{13} \cos a^\circ = 2$<br>$\sqrt{13} \sin a^\circ = 3$<br><br>$\tan a^\circ = \frac{3}{2}$<br>$a = 56 \cdot 3$<br><br>$\sqrt{13} \cos(x + 56 \cdot 3)^\circ$ | $k \cos x^\circ \cos a^\circ - k \sin x^\circ \sin a^\circ$<br>$\cos a^\circ = 2$<br>$\sin a^\circ = 3$<br><br>$\tan a^\circ = \frac{3}{2}$<br>$a = 56 \cdot 3$<br><br>$\sqrt{13} \cos(x + 56 \cdot 3)^\circ$ | $\cos x^\circ \cos a^\circ - \sin x^\circ \sin a^\circ$<br>$\cos a^\circ = 2$<br>$\sin a^\circ = 3$<br><br>$k = \sqrt{13}$<br>$\tan a^\circ = \frac{3}{2}$<br>$a = 56 \cdot 3$<br><br>$\sqrt{13} \cos(x + 56 \cdot 3)^\circ$ |
| • <sup>1</sup><br>• <sup>2</sup><br>• <sup>3</sup><br>• <sup>4</sup>                                                                                               | • <sup>1</sup><br>• <sup>2</sup><br>Not consistent with equations at • <sup>2</sup> .<br>• <sup>3</sup><br>• <sup>4</sup>                                                                                     | • <sup>1</sup><br>• <sup>2</sup> 2<br>• <sup>3</sup><br>• <sup>4</sup>                                                                                                                                                       |

| Question                                                                                                                                                                                                                                                                                                                 |  | Generic scheme                                                                                                                                                                                                                                                                                                               |  | Illustrative scheme                                                                                                                                                                                                                                                                                        |  | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------|
| <b>Candidate D</b> - errors at • <sup>2</sup><br>$k \cos x^\circ \cos a^\circ - k \sin x^\circ \sin a^\circ$<br>• <sup>1</sup> ✓<br>$k \cos a^\circ = 3$<br>$k \sin a^\circ = 2$ • <sup>2</sup> ✗<br>$\tan a^\circ = \frac{2}{3}$<br>$a = 33.7$<br>$\sqrt{13} \cos(x+33.7)^\circ$ • <sup>3</sup> ✓<br>• <sup>4</sup> ✓ 1 |  | <b>Candidate E</b> - errors at • <sup>2</sup><br>$k \cos x^\circ \cos a^\circ - k \sin x^\circ \sin a^\circ$<br>• <sup>1</sup> ✓<br>$k \cos a^\circ = 2$<br>$k \sin a^\circ = -3$ • <sup>2</sup> ✗<br>$\tan a^\circ = -\frac{3}{2}$<br>$a = 303.7$<br>$\sqrt{13} \cos(x+303.7)^\circ$ • <sup>3</sup> ✓<br>• <sup>4</sup> ✓ 1 |  | <b>Candidate F</b> - use of $x$<br>$k \cos x^\circ \cos a^\circ - k \sin x^\circ \sin a^\circ$<br>• <sup>1</sup> ✓<br>$k \cos x^\circ = 2$<br>$k \sin x^\circ = 3$ • <sup>2</sup> ✗<br>$\tan a^\circ = \frac{3}{2}$<br>$x = 56.3$<br>$\sqrt{13} \cos(x+56.3)^\circ$ • <sup>3</sup> ✓<br>• <sup>4</sup> ✓ 1 |  |          |
| <b>Candidate G</b><br>$k \cos A \cos B - k \sin A \sin B$<br>• <sup>1</sup> ✗<br>$k \cos A^\circ = 2$<br>$k \sin A^\circ = 3$ • <sup>2</sup> ✗<br>$\tan A^\circ = \frac{3}{2}$<br>$a = 56.3$<br>$\sqrt{13} \cos(x+56.3)^\circ$ • <sup>3</sup> ✓<br>• <sup>4</sup> ✓ 1                                                    |  |                                                                                                                                                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                                                                            |  |          |
| (b)                                                                                                                                                                                                                                                                                                                      |  | • <sup>5</sup> link to (a)<br><br>• <sup>6</sup> solve for $x+a$<br><br>• <sup>7</sup> solve for $x$                                                                                                                                                                                                                         |  | • <sup>5</sup> $\sqrt{13} \cos(x+56.3\dots)^\circ = 3$<br><br>• <sup>6</sup> 33.69...(393.69...)      • <sup>7</sup> 326.31...<br><br>• <sup>7</sup> 337.38...      270                                                                                                                                    |  | 3        |
| Notes:                                                                                                                                                                                                                                                                                                                   |  |                                                                                                                                                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                                                                            |  |          |
| 11. Do not penalise working which rounds to 34, 326, 394 leading to 270 and 337.                                                                                                                                                                                                                                         |  |                                                                                                                                                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                                                                            |  |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                             |  |                                                                                                                                                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                                                                            |  |          |
|                                                                                                                                                                                                                                                                                                                          |  |                                                                                                                                                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                                                                            |  |          |

| Question                                                                                                                                                                                                                                                       |     |  | Generic scheme                                                                                                                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                      | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.                                                                                                                                                                                                                                                             | (a) |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> identify common factor</li> <li>•<sup>2</sup> complete the square</li> <li>•<sup>3</sup> process for <math>r</math> and write in required form</li> </ul>                                    | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>-6(x^2 - 4x \dots)</math> stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>-6(x-2)^2 \dots</math></li> <li>•<sup>3</sup> <math>-6(x-2)^2 - 1</math></li> </ul> | 3        |
|                                                                                                                                                                                                                                                                |     |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> expand completed square form</li> <li>•<sup>2</sup> equate coefficients</li> <li>•<sup>3</sup> process for <math>q</math> and <math>r</math> and write in required form</li> </ul>           | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>px^2 + 2pqx + pq^2 + r</math></li> <li>•<sup>2</sup> <math>p = -6, 2pq = 24, pq^2 + r = -25</math></li> <li>•<sup>3</sup> <math>-6(x-2)^2 - 1</math></li> </ul>               |          |
| <b>Notes:</b> <ol style="list-style-type: none"> <li><math>-6(x-2)^2 - 1</math> with no working gains •<sup>1</sup> and •<sup>2</sup> only. However, see Candidate E.</li> <li>•<sup>3</sup> is not available in cases where <math>p &gt; 0</math>.</li> </ol> |     |  |                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                          |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                            |     |  |                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                          |          |
| <b>Candidate A</b><br>$-6(x^2 - 4) - 25$<br>$-6((x-2)^2 - 4) - 25$ • <sup>1</sup> ✓    • <sup>2</sup> ✓<br>$-6(x-2)^2 - 1$ • <sup>3</sup> ✓<br>See the exception to general marking principle (h)                                                              |     |  | <b>Candidate B</b><br>$px^2 + 2pqx + pq^2 + r$ • <sup>1</sup> ✓<br>$p = -6, 2pq = 24, pq^2 + r = -25$ • <sup>2</sup> ✓<br>$q = -2, r = -1$ • <sup>3</sup> ^<br><div>             •<sup>3</sup> is lost as answer is not in completed square form           </div> |                                                                                                                                                                                                                                                          |          |
| <b>Candidate C</b><br>$-6(x^2 + 24x) - 25$ • <sup>1</sup> ✗<br>$-6((x+12)^2 - 144) - 25$ • <sup>2</sup> ✓ 1<br>$-6(x+12)^2 + 839$ • <sup>3</sup> ✓ 1                                                                                                           |     |  | <b>Candidate D</b><br>$-6((x+12)^2 - 144) - 25$ • <sup>1</sup> ^    • <sup>2</sup> ✗<br>$-6(x+12)^2 + 839$ • <sup>3</sup> ✓ 1                                                                                                                                     |                                                                                                                                                                                                                                                          |          |
| <b>Candidate E</b><br>$-6(x-2)^2 - 1$<br>Check: $= -6(x^2 - 4x + 4) - 1$<br>$= -6x^2 + 24x - 24 - 1$<br>$= -6x^2 + 24x - 25$<br>Award 3/3                                                                                                                      |     |  | <b>Candidate F</b><br>$-6x^2 + 24x - 25$<br>$= 6x^2 - 24x + 25$ • <sup>1</sup> ✗<br>$= 6(x^2 - 4x \dots)$<br>$= 6(x-2)^2 \dots$ • <sup>2</sup> ✓ 1<br>$= -6(x-2)^2 \dots$ • <sup>3</sup> ✗                                                                        |                                                                                                                                                                                                                                                          |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | Generic scheme                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                          | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (b) |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>4</sup> differentiate</li> <li>•<sup>5</sup> link with (a) and identify sign of <math>(x-2)^2</math></li> <li>•<sup>6</sup> communicate reason</li> </ul> | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>-6x^2 + 24x - 25</math></li> <li>•<sup>5</sup> <math>f'(x) = -6(x-2)^2 - 1</math> and <math>(x-2)^2 \geq 0 \forall x</math></li> <li>•<sup>6</sup> eg <math>\therefore -6(x-2)^2 - 1 &lt; 0 \forall x</math><br/><math>\Rightarrow</math> always strictly decreasing</li> </ul>                                                                   | 3        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>4</sup> differentiate</li> <li>•<sup>5</sup> identify maximum value of <math>f'(x)</math></li> <li>•<sup>6</sup> communicate reason</li> </ul>            | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>-6x^2 + 24x - 25</math></li> <li>•<sup>5</sup> 'maximum value is <math>-1</math>' or annotated sketch including <math>x</math>-axis</li> <li>•<sup>6</sup> <math>-1 &lt; 0</math> or 'graph lies below <math>x</math>-axis'<br/><math>\therefore f'(x) &lt; 0 \forall x</math><br/><math>\Rightarrow</math> always strictly decreasing</li> </ul> |          |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <p>3. In Method 1, do not penalise <math>(x-2)^2 &gt; 0</math> or the omission of <math>f'(x)</math> at •<sup>5</sup>.</p> <p>4. In Method 1, accept <math>-6(x-2)^2 \leq 0</math> or <math>-6(x-2)^2 &lt; 0</math> at •<sup>5</sup>.</p> <p>5. At •<sup>5</sup> communication must be explicitly in terms of the derivative of the given function. Do not accept statements such as '<math>(\text{something})^2 \geq 0</math>', '<del>something squared</del> <math>\geq 0</math>'. However, •<sup>6</sup> is still available.</p> |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Candidate G</b><br>$f'(x) = -6x^2 + 24x - 25$ • <sup>4</sup> ✓<br>$f'(x) = -6(x-2)^2 - 1$ • <sup>5</sup> ^<br>$-6(x-2)^2 - 1 < 0$<br>$\Rightarrow$ strictly decreasing      • <sup>6</sup> ^                                                                                                                                                                                                                                                                                                                                     |     |  |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                              |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  | Generic scheme                                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                 | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (a) |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> equate composite function to <math>x</math></li> <li>•<sup>2</sup> write <math>f(f^{-1}(x))</math> in terms of <math>f^{-1}(x)</math></li> <li>•<sup>3</sup> state inverse function</li> </ul> | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>f(f^{-1}(x)) = x</math></li> <li>•<sup>2</sup> <math>\sqrt[3]{f^{-1}(x)} + 8 = x</math></li> <li>•<sup>3</sup> <math>f^{-1}(x) = (x - 8)^3</math></li> </ul>                                                                                             | 3        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> write as <math>y = f(x)</math> and start to rearrange</li> <li>•<sup>2</sup> express <math>x</math> in terms of <math>y</math></li> <li>•<sup>3</sup> state inverse function</li> </ul>        | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = f(x) \Rightarrow x = f^{-1}(y)</math><br/><math>y - 8 = \sqrt[3]{x}</math></li> <li>•<sup>2</sup> <math>x = (y - 8)^3</math></li> <li>•<sup>3</sup> <math>f^{-1}(y) = (y - 8)^3</math><br/><math>\Rightarrow f^{-1}(x) = (x - 8)^3</math></li> </ul> |          |
| <b>Notes:</b> <ol style="list-style-type: none"> <li>1. In Method 2, accept '<math>y - 8 = \sqrt[3]{x}</math>' without reference to <math>y = f(x) \Rightarrow x = f^{-1}(y)</math> at •<sup>1</sup>.</li> <li>2. In Method 2, accept <math>f^{-1}(x) = (x - 8)^3</math> without reference to <math>f^{-1}(y)</math> at •<sup>3</sup>.</li> <li>3. At •<sup>3</sup> stage, accept <math>f^{-1}</math> written in terms of any dummy variable eg <math>f^{-1}(y) = (y - 8)^3</math>.</li> <li>4. <math>y = (x - 8)^3</math> does not gain •<sup>3</sup>.</li> <li>5. <math>f^{-1}(x) = (x - 8)^3</math> with no working gains 3/3.</li> </ol> |     |  |                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                     |          |

| Question                                                                                                                                                                                                                                                                                                                                                    |     | Generic scheme |                             | Illustrative scheme |                                                                                                                                                                                                | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------|-----------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                |     |                |                             |                     |                                                                                                                                                                                                |          |
| Candidate A - multiple expressions for $y = f(x)$<br>$f(x) = \sqrt[3]{x} + 8$<br>$y = \sqrt[3]{x} + 8$<br>$y - 8 = \sqrt[3]{x}$<br>$x = (y - 8)^3$<br>$y = (x - 8)^3$<br>$f^{-1}(x) = (x - 8)^3$                                                                                                                                                            |     |                | Award 2/3                   |                     | Candidate B - multiple expressions for $y = f(x)$<br>$f(x) = \sqrt[3]{x} + 8$<br>$y = \sqrt[3]{x} + 8$<br>$x = \sqrt[3]{y} + 8$<br>$y = (x - 8)^3$<br>$f^{-1}(x) = (x - 8)^3$<br><br>Award 2/3 |          |
| Candidate C - BEWARE<br>$f'(x) = \dots$                                                                                                                                                                                                                                                                                                                     |     |                | • <sup>3</sup> ✗            |                     | Candidate D<br>$f^{-1}(x) = x - 8^3$<br>with no working<br><br>Award 0/3                                                                                                                       |          |
| Candidate E<br>$x \rightarrow \sqrt[3]{x} \rightarrow \sqrt[3]{x} + 8 = f(x)$<br>$\sqrt[3]{\phantom{x}} \rightarrow +8$<br>$\therefore -8 \rightarrow (\phantom{x})^3$<br>$(x - 8)^3$<br>$f^{-1}(x) = (x - 8)^3$<br><br>• <sup>1</sup> ✓<br>• <sup>2</sup> ✓<br>• <sup>3</sup> ✓<br><div>awarded for knowing to perform inverse operations in reverse</div> |     |                |                             |                     |                                                                                                                                                                                                |          |
|                                                                                                                                                                                                                                                                                                                                                             | (b) |                | • <sup>4</sup> state domain |                     | • <sup>4</sup> $9 \leq x \leq 18, x \in \mathbb{R}$                                                                                                                                            | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                      |     |                |                             |                     |                                                                                                                                                                                                |          |
| 1. Do not penalise the omission of $x \in \mathbb{R}$ .                                                                                                                                                                                                                                                                                                     |     |                |                             |                     |                                                                                                                                                                                                |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                |     |                |                             |                     |                                                                                                                                                                                                |          |
|                                                                                                                                                                                                                                                                                                                                                             |     |                |                             |                     |                                                                                                                                                                                                |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  | Generic scheme                                                                                                                                        | Illustrative scheme                                                                                                                                                                          | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (a) |  | • <sup>1</sup> identify initial power                                                                                                                 | • <sup>1</sup> 120                                                                                                                                                                           | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (b) |  | • <sup>2</sup> interpret information<br>• <sup>3</sup> process equation<br>• <sup>4</sup> write in logarithmic form<br>• <sup>5</sup> process for $t$ | • <sup>2</sup> $102 = 120e^{-0.0079t}$<br>stated or implied by • <sup>3</sup><br>• <sup>3</sup> $e^{-0.0079t} = 0.85$<br>• <sup>4</sup> $\log_e 0.85 = -0.0079t$<br>• <sup>5</sup> 20.572... | 4        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| 1. Candidates who interpret 15% incorrectly do not gain • <sup>2</sup> , but • <sup>3</sup> , • <sup>4</sup> and • <sup>5</sup> are still available. See Candidate E.<br>2. • <sup>3</sup> may be implied by • <sup>4</sup> .<br>3. Any base may be used at • <sup>4</sup> stage. See Candidate A.<br>4. Accept $\ln 0.85 = -0.0079t \ln e$ for • <sup>4</sup> .<br>5. Accept 20.57 or 20.6 at • <sup>5</sup> .<br>6. The calculation at • <sup>5</sup> must follow from the valid use of exponentials and logarithms at • <sup>3</sup> and • <sup>4</sup> .<br>7. For candidates who take an iterative approach to arrive at $t = 20.6$ award 1/4.<br>However, if, in the iterations $P_t$ is evaluated for $t = 20.55$ and $t = 20.65$ then award 4/4. |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| Candidate A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  | Candidate B                                                                                                                                           |                                                                                                                                                                                              |          |
| $102 = 120e^{-0.0079t}$<br>$e^{-0.0079t} = 0.85$<br>$\log_{10} 0.85 = -0.0079t \log_{10} e$<br>$20.6$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | $102 = 120e^{-0.0079t}$<br>$e^{-0.0079t} = 0.85$<br>$t = 20.6$                                                                                        |                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| • <sup>2</sup> ✓<br>• <sup>3</sup> ✓<br>• <sup>4</sup> ✓<br>• <sup>5</sup> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  | • <sup>2</sup> ✓<br>• <sup>3</sup> ✓<br>• <sup>4</sup> ^   • <sup>5</sup> <span style="border: 1px solid red; padding: 0 2px;">1</span>               |                                                                                                                                                                                              |          |
| Candidate C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  | Candidate D                                                                                                                                           |                                                                                                                                                                                              |          |
| $\log_e 0.85 = -0.0079t$<br>$t = 20.6$ years<br>$t = 20$ years 6 months                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  | $\log_e 0.85 = -0.0079t$<br>$t = 20$ years 6 months                                                                                                   |                                                                                                                                                                                              |          |
| • <sup>4</sup> ✓<br>• <sup>5</sup> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  | • <sup>4</sup> ✓<br>• <sup>5</sup> ✗                                                                                                                  |                                                                                                                                                                                              |          |
| <div style="border: 1px solid black; border-radius: 10px; padding: 10px; display: inline-block;">             Incorrect conversion<br/>             subsequent to answer<br/>             is not penalised           </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| Candidate E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| $15 = 100e^{-0.0079t}$<br>$e^{-0.0079t} = 0.15$<br>$\log_e 0.15 = -0.0079t$<br>$240.1...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |
| • <sup>2</sup> ✗<br>• <sup>3</sup> <span style="border: 1px solid red; padding: 0 2px;">1</span><br>• <sup>4</sup> <span style="border: 1px solid red; padding: 0 2px;">1</span><br>• <sup>5</sup> <span style="border: 1px solid red; padding: 0 2px;">1</span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |  |                                                                                                                                                       |                                                                                                                                                                                              |          |



| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |  | Generic scheme                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (a) |  | <p>•<sup>1</sup> use <math>-3</math> in synthetic division or in evaluation of quartic</p> <p>•<sup>2</sup> complete division/evaluation and interpret result</p> | <p>•<sup>1</sup> <math display="block">\begin{array}{r rrrrr} -3 &amp; 3 &amp; 10 &amp; 1 &amp; -8 &amp; -6 \\ &amp; &amp; -9 &amp; -3 &amp; 6 &amp; 6 \\ \hline &amp; 3 &amp; 1 &amp; -2 &amp; -2 &amp; 0 \end{array}</math></p> <p>or</p> $3 \times (-3)^4 + 10 \times (-3)^3 + (-3)^2 - 8 \times (-3) - 6$ <p>•<sup>2</sup> <math display="block">\begin{array}{r rrrrr} -3 &amp; 3 &amp; 10 &amp; 1 &amp; -8 &amp; -6 \\ &amp; &amp; -9 &amp; -3 &amp; 6 &amp; 6 \\ \hline &amp; 3 &amp; 1 &amp; -2 &amp; -2 &amp; 0 \end{array}</math></p> <p>Remainder = 0 <math>\therefore (x+3)</math> is a factor<br/>or <math>f(-3) = 0 \therefore (x+3)</math> is a factor</p> | 2        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <p>1. Communication at •<sup>2</sup> must be consistent with working at that stage ie a candidate's working must arrive legitimately at 0 before •<sup>2</sup> can be awarded.</p> <p>2. Accept any of the following for •<sup>2</sup>:</p> <ul style="list-style-type: none"> <li>• '<math>f(-3) = 0</math> so <math>(x+3)</math> is a factor'</li> <li>• 'since remainder = 0, it is a factor'</li> <li>• the '0' from any method linked to the word 'factor' by 'so', 'hence', <math>\therefore</math>, <math>\rightarrow</math>, <math>\Rightarrow</math> etc.</li> </ul> <p>3. Do not accept any of the following for •<sup>2</sup>:</p> <ul style="list-style-type: none"> <li>• double underlining the '0' or boxing the '0' without comment</li> <li>• '<math>x = -3</math> is a factor', '... is a root'</li> <li>• the word 'factor' only, with no link.</li> </ul> |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  |                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |

| Question |     | Generic scheme                                                                                                                                                                                                                                                                                             | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Max mark |
|----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | (b) | <ul style="list-style-type: none"> <li>•<sup>3</sup> identify cubic and attempt to factorise</li> <li>•<sup>4</sup> find second factor</li> <li>•<sup>5</sup> identify quadratic</li> <li>•<sup>6</sup> evaluate discriminant</li> <li>•<sup>7</sup> interpret discriminant and factorise fully</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>3</sup> eg <math display="block">\begin{array}{r rrrr} &amp; 3 &amp; 1 &amp; -2 &amp; -2 \\ &amp; \dots &amp; \dots &amp; \dots &amp; \dots \end{array}</math></li> <li>•<sup>4</sup> eg <math display="block">\begin{array}{r rrrr} 1 &amp; 3 &amp; 1 &amp; -2 &amp; -2 \\ &amp; 3 &amp; 4 &amp; 2 &amp; \\ \hline &amp; 3 &amp; 4 &amp; 2 &amp; 0 \end{array}</math></li> <li>leading to <math>(x-1)</math></li> <li>•<sup>5</sup> <math>3x^2 + 4x + 2</math></li> <li>•<sup>6</sup> <math>-8</math></li> <li>•<sup>7</sup> since <math>-8 &lt; 0</math>, quadratic has no (real) factors leading to <math>(x+3)(x-1)(3x^2 + 4x + 2)</math></li> </ul> | 5        |

#### Notes:

- Candidates who arrive at  $(x+3)(x-1)(3x^2 + 4x + 2)$  by using algebraic long division or by inspection gain •<sup>3</sup>, •<sup>4</sup> and •<sup>5</sup>.
- Evidence for •<sup>6</sup> may appear in the quadratic formula.
- Accept ' $-8 < 0$  so no real roots' with the fully factorised quartic for •<sup>7</sup>:
- Do not accept any of the following for •<sup>7</sup>:
  - $(x+3)(x-1)(3x^2 + 4x + 2)$  does not factorise
  - $(x+3)(x-1)(\dots \dots)(\dots \dots)$  cannot factorise further.
- Accept  $(x+3)(x-1)3x^2 + 4x + 2$ , with a valid reason for •<sup>7</sup>.
- Where the quadratic factor obtained at •<sup>5</sup> can be factorised, •<sup>6</sup> and •<sup>7</sup> are not available.

#### Commonly Observed Responses:

| Candidate A                 |                    | Candidate B                 |                  |
|-----------------------------|--------------------|-----------------------------|------------------|
| $(x+3)(x-1)(3x^2 + 4x + 2)$ | • <sup>5</sup> ✓   | $(x+3)(x-1)(3x^2 + 4x + 2)$ | • <sup>5</sup> ✓ |
| $b^2 - 4ac = 16 - 24 < 0$   | • <sup>6</sup> ^   | $b^2 - 4ac < 0$             | • <sup>6</sup> ^ |
| so does not factorise       | • <sup>7</sup> ✓ 1 | so does not factorise       | • <sup>7</sup> ^ |

| Question                                                                                                                                                                                                                                                                          |     |  | Generic scheme                                                                                                                                                                                                                                                                                                                                        | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.                                                                                                                                                                                                                                                                               | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> express <math>A</math> in terms of <math>x</math> and <math>h</math></li> <li>•<sup>2</sup> express height in terms of <math>x</math></li> <li>•<sup>3</sup> substitute for <math>h</math> and complete proof</li> </ul>                                                                         | <ul style="list-style-type: none"> <li>•<sup>1</sup> (<math>A=</math>) <math>16x^2 + 16xh</math></li> <li>•<sup>2</sup> <math>h = \frac{2000}{8x^2}</math></li> <li>•<sup>3</sup> <math>A = 16x^2 + 16x \times \frac{2000}{8x^2}</math><br/>leading to <math>A = 16x^2 + \frac{4000}{x}</math></li> </ul>                                                                                                                                                             | 3        |
| Notes:                                                                                                                                                                                                                                                                            |     |  |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| 1. At • <sup>1</sup> accept any unsimplified form of $16x^2 + 16xh$ .<br>2. The substitution for $h$ at • <sup>3</sup> must be clearly shown for • <sup>3</sup> to be available.<br>3. For candidates who omit some of the surfaces of the box, only • <sup>2</sup> is available. |     |  |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                      |     |  |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
|                                                                                                                                                                                                                                                                                   | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> express <math>A</math> in differentiable form</li> <li>•<sup>5</sup> differentiate</li> <li>•<sup>6</sup> equate expression for derivative to 0</li> <li>•<sup>7</sup> process for <math>x</math></li> <li>•<sup>8</sup> verify nature</li> <li>•<sup>9</sup> evaluate <math>A</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>16x^2 + 4000x^{-1}</math></li> <li>•<sup>5</sup> <math>32x - 4000x^{-2}</math></li> <li>•<sup>6</sup> <math>32x - 4000x^{-2} = 0</math></li> <li>•<sup>7</sup> 5</li> <li>•<sup>8</sup> table of signs for a derivative (see below) <math>\therefore</math> minimum<br/>or <math>A''(x) = 96 &gt; 0 \Rightarrow</math> minimum</li> <li>•<sup>9</sup> <math>A = 1200</math> or min value = 1200</li> </ul> | 6        |

## Notes:

- For a numerical approach award 0/6.
- <sup>6</sup> can be awarded for  $32x = 4000x^{-2}$ .
- For candidates who integrate any term at the •<sup>5</sup> stage, only •<sup>6</sup> is available on follow through for setting their 'derivative' to 0.
- <sup>7</sup>, •<sup>8</sup> and •<sup>9</sup> are only available for working with a derivative which contains an index  $\leq -2$ .
- $\sqrt[3]{\frac{4000}{32}}$  must be simplified at •<sup>7</sup> or •<sup>8</sup> for •<sup>7</sup> to be awarded.
- <sup>8</sup> is not available to candidates who consider a value of  $x \leq 0$  in the neighbourhood of 5.
- <sup>9</sup> is still available in cases where a candidate's table of signs does not lead legitimately to a minimum at •<sup>8</sup>.
- <sup>8</sup> and •<sup>9</sup> are not available to candidates who state that the minimum exists at a negative value of  $x$ . See Candidates C and D.

For the table of signs for a derivative, accept:

| $x$            | $5^-$ | $5$ | $5^+$ |
|----------------|-------|-----|-------|
| $A'(x)$        | -     | 0   | +     |
| Shape or slope | \     | -   | /     |

| $x$            | $\rightarrow$ | $5$ | $\rightarrow$ |
|----------------|---------------|-----|---------------|
| $A'(x)$        | -             | 0   | +             |
| Shape or slope | \             | -   | /             |

Arrows are taken to mean  
'in the neighbourhood of'

| $x$            | $a$ | $5$ | $b$ |
|----------------|-----|-----|-----|
| $A'(x)$        | -   | 0   | +   |
| Shape or slope | \   | -   | /   |

Where  $0 < a < 5$  and  $b > 5$

- For this question do not penalise the omission of ' $x$ ' or the word 'shape'/'slope'.
- Stating values of  $A'(x)$  in the table is an acceptable alternative to writing '+' or '-' signs. Values must be checked for accuracy.
- The only acceptable variations of  $A'(x)$  are:  $A'$ ,  $\frac{dA}{dx}$  and  $32x - 4000x^{-2}$ .

## Commonly Observed Responses:

**Candidate A** - differentiating over multiple lines

$$A'(x) = 32x + 4000x^{-1} \quad \bullet^4 \checkmark$$

$$A'(x) = 32x - 4000x^{-2} \quad \bullet^5 \times$$

$$32x - 4000x^{-2} = 0 \quad \bullet^6 \checkmark 1$$

**Candidate B** - differentiating over multiple lines

$$A = 16x^2 + 4000x^{-1} \quad \bullet^4 \checkmark$$

$$A'(x) = 32x + 4000x^{-1} \quad \bullet^5 \times$$

$$A'(x) = 32x - 4000x^{-2} \quad \bullet^6 \checkmark 1$$

$$32x - 4000x^{-2} = 0$$

**Candidate C** - only considers 5

$$A = 16x^2 + 4000x^{-1} \quad \bullet^4 \checkmark$$

$$A' = 32x - 4000x^{-2} = 0 \quad \bullet^5 \checkmark \quad \bullet^6 \checkmark$$

$$x = \pm 5 \quad \bullet^7 \times$$

| $x$  | $\rightarrow$ | $5$ | $\rightarrow$ |
|------|---------------|-----|---------------|
| $A'$ | -             | 0   | +             |
|      | \             | -   | /             |

$\therefore$  minimum  $\bullet^8 \checkmark 1$

$A = 1200$  or min value = 1200  $\bullet^9 \checkmark 1$

**Candidate D** - considers 5 and negative 5 in separate tables

$$A = 16x^2 + 4000x^{-1} \quad \bullet^4 \checkmark$$

$$A' = 32x - 4000x^{-2} = 0 \quad \bullet^5 \checkmark \quad \bullet^6 \checkmark$$

$$x = \pm 5 \quad \bullet^7 \times$$

| $x$  | $\rightarrow$ | $5$ | $\rightarrow$ |
|------|---------------|-----|---------------|
| $A'$ | -             | 0   | +             |
|      | \             | -   | /             |

| $x$  | $\rightarrow$ | $-5$ | $\rightarrow$ |
|------|---------------|------|---------------|
| $A'$ | -             | 0    | +             |
|      | /             | -    | \             |

$\therefore$  minimum when  $x = 5$   $\bullet^8 \checkmark 1$

$A = 1200$  or min value = 1200  $\bullet^9 \checkmark 1$

Ignore incorrect  
working in  
second table

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Max mark |
|----------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 12.      |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> state linear equation</li> <li>•<sup>2</sup> introduce logs</li> <li>•<sup>3</sup> use laws of logs</li> <li>•<sup>4</sup> use laws of logs</li> <li>•<sup>5</sup> state <math>a</math> and <math>b</math></li> </ul>                          | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_4 y = 3x - 1</math></li> <li>•<sup>2</sup> <math>\log_4 y = 3x \log_4 4 - \log_4 4</math></li> <li>•<sup>3</sup> <math>\log_4 y = \log_4 4^{3x} - \log_4 4</math></li> <li>•<sup>4</sup> <math>\log_4 y = \log_4 \left( \frac{4^{3x}}{4} \right)</math> or<br/><math>\log_4 y = \log_4 4^{-1} 4^{3x}</math></li> <li>•<sup>5</sup> <math>a = \frac{1}{4}, b = 64</math></li> </ul>         | 5        |
|          |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> state linear equation</li> <li>•<sup>2</sup> convert to exponential form</li> <li>•<sup>3</sup> use laws of indices</li> <li>•<sup>4</sup> state <math>a</math></li> <li>•<sup>5</sup> state <math>b</math></li> </ul>                         | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_4 y = 3x - 1</math></li> <li>•<sup>2</sup> <math>y = 4^{3x-1}</math></li> <li>•<sup>3</sup> <math>y = 4^{-1} 4^{3x}</math></li> <li>•<sup>4</sup> <math>a = \frac{1}{4}</math></li> <li>•<sup>5</sup> <math>b = 64</math></li> </ul>                                                                                                                                                       | 5        |
|          |  |  | <b>Method 3</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> introduce logs to <math>y = ab^x</math></li> <li>•<sup>2</sup> use laws of logs</li> <li>•<sup>3</sup> interpret intercept</li> <li>•<sup>4</sup> interpret gradient</li> <li>•<sup>5</sup> state <math>a</math> and <math>b</math></li> </ul> | <b>Method 3</b><br>The equations at • <sup>1</sup> , • <sup>2</sup> , • <sup>3</sup> and • <sup>4</sup> must be stated explicitly. <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_4 y = \log_4 ab^x</math></li> <li>•<sup>2</sup> <math>\log_4 y = \log_4 a + x \log_4 b</math></li> <li>•<sup>3</sup> <math>-1 = \log_4 a</math></li> <li>•<sup>4</sup> <math>3 = \log_4 b</math></li> <li>•<sup>5</sup> <math>a = \frac{1}{4}, b = 64</math></li> </ul> | 5        |

| Question                                                                                                                                                                                                                                                                                                                                                                                 |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                        | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                                                                                                                                                                                                                                                                                                                                                                          |  |  | <b>Method 4</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> interpret point on log graph</li> <li>•<sup>2</sup> convert from log to exponential form</li> <li>•<sup>3</sup> interpret point and convert</li> <li>•<sup>4</sup> substitute into <math>y=ab^x</math> and evaluate <math>a</math></li> <li>•<sup>5</sup> substitute other point into <math>y=ab^x</math> and evaluate <math>b</math></li> </ul> | <b>Method 4</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>x=3</math> and <math>\log_4 y=8</math></li> <li>•<sup>2</sup> <math>x=3</math> and <math>y=4^8</math></li> <li>•<sup>3</sup> <math>x=0</math> and <math>\log_4 y=-1</math><br/><math>x=0</math> and <math>y=4^{-1}</math></li> <li>•<sup>4</sup> <math>4^{-1} = ab^0 \Rightarrow a = \frac{1}{4}</math></li> <li>•<sup>5</sup> <math>4^8 = \frac{1}{4}b^3 \Rightarrow b = 64</math></li> </ul> | 5        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| 1. In any method, marks may only be awarded within a valid strategy using $y=ab^x$ .<br>2. Accept $y = \frac{1}{4} \cdot 64^x$ for • <sup>5</sup> .<br>3. Markers must identify the method which best matches the candidates approach; they must not mix and match between methods.<br>4. Penalise the omission of base 4 at most once in any method.<br>5. Do not accept $a = 4^{-1}$ . |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                      |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                                                                                                                          |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                        | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                   | Max mark |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 13.      |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> interpret information given</li> <li>•<sup>2</sup> integrate any two terms</li> <li>•<sup>3</sup> complete integration</li> <li>•<sup>4</sup> interpret information given and substitute</li> <li>•<sup>5</sup> process for <math>c</math> and state expression for <math>f(x)</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>f'(x) = 3x^2 - 16x + 11</math> or <math>f(x) = \int (3x^2 - 16x + 11) dx</math></li> <li>•<sup>2</sup> eg <math>\frac{3x^3}{3} - \frac{16x^2}{2} \dots</math></li> <li>•<sup>3</sup> <math>\dots + 11x + c</math></li> <li>•<sup>4</sup> <math>0 = 7^3 - 8 \times 7^2 + 11 \times 7 + c</math></li> <li>•<sup>5</sup> <math>f(x) = x^3 - 8x^2 + 11x - 28</math></li> </ul> | 5        |

#### Notes:

1. For candidates who make no attempt to integrate to find  $f(x)$  award 0/5.
2. Do not penalise the omission of  $f(x)$  or  $dx$  or the appearance of  $+c$  at •<sup>1</sup>.
3. If any two terms have been integrated correctly •<sup>1</sup> may be implied by •<sup>2</sup>.
4. For candidates who omit  $+c$ , only •<sup>1</sup> and •<sup>2</sup> are available.
5. For candidates who differentiate **any** term, •<sup>3</sup> •<sup>4</sup> and •<sup>5</sup> are not available.
6. Candidates must attempt to integrate both terms containing  $x$  for •<sup>4</sup> and •<sup>5</sup> to be available. See Candidate B.
7. Accept  $y = x^3 - 8x^2 + 11x - 28$  at •<sup>5</sup> since  $y = f(x)$  is defined in the question.
8. Candidates must simplify coefficients in **their** final line of working for the last mark available in that line of working to be awarded.

#### Commonly Observed Responses:

##### Candidate A - incomplete substitution

$$f(x) = x^3 - 8x^2 + 11x + c \quad \bullet^1 \checkmark \quad \bullet^2 \checkmark \quad \bullet^3 \checkmark$$

$$f(x) = 7^3 - 8 \times 7^2 + 11 \times 7 + c \quad \bullet^4 \wedge$$

$$c = -28$$

$$f(x) = x^3 - 8x^2 + 11x - 28 \quad \bullet^5 \boxed{\checkmark 1}$$

##### Candidate B - partial integration

$$f(x) = x^3 - 8x^2 + 11 + c \quad \bullet^1 \checkmark \quad \bullet^2 \checkmark \quad \bullet^3 \times$$

$$0 = 7^3 - 8 \times 7^2 + 11 + c \quad \bullet^4 \boxed{\checkmark 1}$$

$$c = 38$$

$$f(x) = x^3 - 8x^2 + 49 \quad \bullet^5 \boxed{\checkmark 1}$$

| Question |  |  | Generic scheme                                                                                                                                                                                                                               | Illustrative scheme                                                                                                                                                                                                                                                                                    | Max mark |
|----------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 14.      |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> expand</li> <li>•<sup>2</sup> evaluate <math>\mathbf{u.u}</math></li> <li>•<sup>3</sup> determine equation in <math>\cos \theta</math></li> <li>•<sup>4</sup> evaluate angle</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\mathbf{uu+uv}</math></li> <li>•<sup>2</sup> 16</li> <li>•<sup>3</sup> <math>20\cos\theta=5</math> or <math>\cos\theta=\frac{5}{20}</math></li> <li>•<sup>4</sup> <math>75.5\dots^\circ</math> or <math>1.31\dots</math> radians</li> </ul> | 4        |

#### Notes:

1. Do not accept  $\mathbf{u}^2$  for •<sup>1</sup>, however •<sup>2</sup>, •<sup>3</sup> and •<sup>4</sup> are still available.
2. Where there is no evidence for •<sup>1</sup>, then •<sup>2</sup>, •<sup>3</sup> and •<sup>4</sup> are not available, however see Candidates C and D.
3. Where candidates use  $|\mathbf{u}| \neq 4$ , then •<sup>3</sup> and •<sup>4</sup> are not available.
4. Where there is no evidence of using  $|\mathbf{u}|^2$ , •<sup>3</sup> is not available. See Candidate A.
5. Do not penalise omission of units in final answer.
6. Ignore the appearance of  $284.5^\circ$ .
7. Accept answers which round to  $76^\circ$  or 1.3 radians.

#### Commonly Observed Responses:

|                                                                                                                                                                                                              |                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate A</b><br>$\mathbf{u.(u+v)=u.u+u.v}$ • <sup>1</sup> ✓<br>$4+20\cos\theta=21$ • <sup>2</sup> ✗<br>$\cos\theta=\frac{17}{20}$ • <sup>3</sup> ✓ 2<br>$\theta=31.7\dots^\circ$ • <sup>4</sup> ✓ 1    | <b>Candidate B</b><br>$16+\mathbf{u.v}=21$ • <sup>1</sup> ✓    • <sup>2</sup> ✓<br>$\mathbf{u.v}=5$<br>$\cos\theta=\frac{5}{20}$ • <sup>3</sup> ✓<br>$\theta=75.5^\circ$ • <sup>4</sup> ✓ |
| <b>Candidate C - missing working</b><br>$\mathbf{u.u}=16$ • <sup>2</sup> ✓<br>$\mathbf{u.v}=21-16$<br>$\cos\theta=\frac{5}{20}$ • <sup>1</sup> ✓    • <sup>3</sup> ✓<br>$\theta=75.5^\circ$ • <sup>4</sup> ✓ | <b>Candidate D - missing working</b><br>$21-16=5$ • <sup>1</sup> ^<br>$\cos\theta=\frac{5}{20}$ • <sup>2</sup> ✓    • <sup>3</sup> ✓<br>$\theta=75.5^\circ$ • <sup>4</sup> ✓              |

| Question                                                                                                                                                                                                                                                                                                                   |     |      | Generic scheme                                                                                                                                                                                 | Illustrative scheme                                                                                                                                                                                                    | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 15.                                                                                                                                                                                                                                                                                                                        | (a) |      | <ul style="list-style-type: none"> <li>•<sup>1</sup> find gradient of radius</li> <li>•<sup>2</sup> state gradient of tangent</li> <li>•<sup>3</sup> state equation of tangent</li> </ul>      | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>-\frac{1}{3}</math></li> <li>•<sup>2</sup> 3</li> <li>•<sup>3</sup> <math>y = 3x - 2</math></li> </ul>                                                      | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                              |     |      |                                                                                                                                                                                                |                                                                                                                                                                                                                        |          |
| 1. Do not accept $y = \frac{3}{1}x - 2$ for • <sup>3</sup> .<br>2. • <sup>3</sup> is only available as a consequence of trying to find and use a perpendicular gradient.<br>3. At • <sup>3</sup> accept, $y - 3x + 2 = 0$ or any other rearrangement of the equation where the constant terms have been simplified.        |     |      |                                                                                                                                                                                                |                                                                                                                                                                                                                        |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                        |     |      |                                                                                                                                                                                                |                                                                                                                                                                                                                        |          |
|                                                                                                                                                                                                                                                                                                                            | (b) | (i)  | • <sup>4</sup> find coordinates of T                                                                                                                                                           | • <sup>4</sup> (0, -2)                                                                                                                                                                                                 | 1        |
|                                                                                                                                                                                                                                                                                                                            |     | (ii) | <ul style="list-style-type: none"> <li>•<sup>5</sup> find midpoint CT</li> <li>•<sup>6</sup> find radius of circle with diameter CT</li> <li>•<sup>7</sup> state equation of circle</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>5</sup> (4, 5)</li> <li>•<sup>6</sup> <math>\sqrt{65}</math> stated or implied by •<sup>7</sup></li> <li>•<sup>7</sup> <math>(x - 4)^2 + (y - 5)^2 = 65</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                              |     |      |                                                                                                                                                                                                |                                                                                                                                                                                                                        |          |
| 4. Answers in part (b)(i) must be consistent with answers from part (a).<br>5. Accept $x = 0$ , $y = -2$ for • <sup>4</sup> .<br>6. $(x - 4)^2 + (y - 5)^2 = (\sqrt{65})^2$ does not gain • <sup>7</sup> .<br>7. • <sup>7</sup> is not available to candidates who use a line other than CT as the diameter of the circle. |     |      |                                                                                                                                                                                                |                                                                                                                                                                                                                        |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                        |     |      |                                                                                                                                                                                                |                                                                                                                                                                                                                        |          |
|                                                                                                                                                                                                                                                                                                                            |     |      |                                                                                                                                                                                                |                                                                                                                                                                                                                        |          |

[END OF MARKING INSTRUCTIONS]